

CHOOSECOLS + XLOOKUP in Excel

1. What is CHOOSECOLS + XLOOKUP?

XLOOKUP finds a value in a list and returns data from the matching row. When it returns several columns at once, CHOOSECOLS can keep only the columns you need. Together they solve the problem of looking up one record and showing just a few of its fields, such as a name and a salary. A Data Analyst uses them to build compact lookup results without writing a separate lookup for every column.

2. Syntax

`=CHOOSECOLS(XLOOKUP(lookup_value, lookup_array, return_array), col_num1, [col_num2], ...)`

Function	Argument	Required?	What it accepts
XLOOKUP	lookup_value	Required	The value to find, such as a cell reference or text in quotes
XLOOKUP	lookup_array	Required	The single column to search (for example A2:A6)
XLOOKUP	return_array	Required	The columns to return from the matching row (for example B2:D6). It needs the same number of rows as lookup_array.
CHOOSECOLS	col_num1	Required	The number of the first column to keep, counted inside return_array. A negative number counts from the right.
CHOOSECOLS	col_num2, ...	Optional	More column numbers, in the order you want them

Because return_array has several columns, XLOOKUP returns the whole matching row. CHOOSECOLS then keeps only the columns you list. XLOOKUP finds an exact match by default.

3. Simple Example

Look up product P004 and return its Name and Sales:

	A	B	C	D	E	F	G	H
1	ID	Name	Region	Sales	Find	Name	Sales	
2	P001	Laptop	North	1200	P004	Monitor	800	
3	P002	Mouse	South	150				
4	P003	Keyboard	North	300				
5	P004	Monitor	East	800				
6	P005	Webcam	South	180				

Formula in G2 only: `=CHOOSECOLS(XLOOKUP(F2, A2:A6, B2:D6), 1, 3)`

XLOOKUP finds P004 and returns its row from B2:D6: Monitor, East, 800. CHOOSECOLS keeps columns 1 and 3 of that row. **Result:**

Monitor and 800

4. Another Example

Look up two employee IDs. One of them does not exist:

	A	B	C	D	E	F	G	H
1	ID	Name	Dept	Salary	Find	Name	Salary	
2	E101	Ayesha	IT	50000	E103	Nabila	61000	
3	E102	Rahim	HR	42000	E105	Not Found		
4	E103	Nabila	IT	61000				
5	E104	Tanvir	Sales	38000				

Formula in G2, copied to G3:

`=IFERROR(CHOOSECOLS(XLOOKUP(F2, A2:A5, B2:D5), 1, 3), "Not Found")`

E103 is found, so you get its Name and Salary. E105 is not in the list, so XLOOKUP gives an error and IFERROR shows "Not Found" instead of #N/A.

5. When to Use It

- Look up a record by ID and show only some of its fields
- Build lookup cards or dashboards that show a few fields for a selected item
- Return several fields with one formula instead of one lookup per column
- Return columns that are not next to each other from a matching row
- Avoid copying columns you do not need into a report

6. Important Things to Remember

- XLOOKUP finds the row. CHOOSECOLS picks the columns from that row.
- Column numbers in CHOOSECOLS count inside return_array, not by sheet column letters. In B2:D6, column 1 is B.
- lookup_array must be a single column, and return_array must have the same number of rows.
- The result spills across columns. The cells to the right must be empty, or Excel shows #SPILL!
- If the lookup value is not found, the result is #N/A. Wrap the whole formula in IFERROR to show a message.
- CHOOSECOLS is available in Microsoft 365 and Excel 2024, but not in Excel 2021 or earlier, even though XLOOKUP is.

7. Common Mistakes

Mistake: Using the sheet column number in CHOOSECOLS, such as 4 for column D in B2:D6, which gives #VALUE!. **Fix:** Count from the first column of return_array.

Mistake: Using ranges with different numbers of rows, such as A2:A6 with B2:D7. **Fix:** Select the same rows for both.

Mistake: Using the if_not_found argument inside XLOOKUP. It returns one value, and CHOOSECOLS then gives #VALUE!. **Fix:** Put IFERROR around the whole formula.

Mistake: Selecting several columns as lookup_array, such as A2:B6. **Fix:** Search in a single column.

8. Data Analyst Tip

You can also put CHOOSECOLS inside XLOOKUP: `=XLOOKUP(F2, A2:A6, CHOOSECOLS(B2:D6, 1, 3))` gives the same result. Use the version you find easier to read. Put the lookup value in a cell and lock the ranges with \$, such as \$A\$2:\$A\$6, so you can copy the formula down.